

AI Use Statement

This project made use of AI coding assistant Anthropic's Claude. This statement describes which parts of the work were assisted by AI, which were not, and what role AI served relative to existing knowledge and skills.

Existing knowledge brought to the project

The following were already understood prior to using AI assistance:

- The function of Kraken2 as a k-mer-based taxonomic classifier and how it differs from other classification approaches.
- The contents of the three Kraken2 databases on the Unity cluster (`kraken2-db`, `PlusPF`, `PrackenDB`) and how they differ in reference scope.
- The biology of *Nephromyces*, including its classification within Apicomplexa and its endosymbiotic relationship.
- Writing and submitting SLURM job scripts, including resource directives (`--cpus-per-task`, `--mem`, `--time`, `--partition`), module loading, and submission via `sbatch`.
- Navigating the cluster filesystem and running Kraken2 against pre-existing databases.
- Working with raw Oxford Nanopore FASTQ data.
- Working with a Github Repository

How AI was used

1. Project structure and reproducibility design. AI was used to lay out the repository structure (`config/`, `scripts/`, `results/`, `paper/`) and to design the project around a single declarative config file rather than hardcoded paths. The pattern of reading database paths from `config/config.yaml` inside `run_pipeline.sh` was suggested by AI and adopted because it made the project portable.

2. Code scaffolding for the Python pipeline. The five Python scripts (`01_parse_kraken.py` through `05_plot_results.py`) were initially scaffolded with AI assistance, including `argparse` boilerplate, YAML loading, error handling for missing files, and the matplotlib plotting code. Each script was then reviewed and modified before use.

3. Pivoting the project scope. When no pre-built Kaiju database was found on the cluster, AI was used to evaluate the tradeoffs between building a Kaiju database from scratch (8–12 hours, ~150 GB) versus pivoting to a Kraken2-only comparison across three databases. The pivot decision and its scientific justification were discussed with other users in the Unity Cluster *Slack* channel and refined through the cluster documentation (<https://docs.unity.rc.umass.edu/documentation/datasets/>).

4. Debugging on the cluster. AI was particularly useful during the debugging phase, which included: SLURM `--chdir` behavior on the URI cluster, missing `pip` in the loaded Python module, an outdated `config.yaml` that hadn't been updated when the project pivoted, and several misleading "No such file or directory" errors. AI helped translate vague error messages into testable diagnostic commands (`cat -A, xxd, file, bash -x`) and walked through the failure modes one at a time.

5. Documentation: Initial drafts of the README, Repository structure and inline docstrings, then revised against the actual project structure and results.

How AI was *not* used

- AI was not used to generate or invent the data, results, or figures. All numbers in the results section come from running Kraken2 against the three real cluster sourced databases on a subset of real Oxford Nanopore reads.
- AI was not used to fabricate biological claims about *Nephromyces*. Statements about its biology and ecological role.
- AI was not used to make the central interpretive judgment of the project, namely, that a high unclassified fraction is *not* evidence that those reads are *Nephromyces*. The cautious framing of the paper is the author's own.
- AI was not used to write or interpret the SLURM directives in `submit_pipeline.sh`. Resource sizing was based on prior experience with Kraken2 jobs on this cluster.

Author responsibility

All code in this repository was reviewed, tested, debugged, and modified by the author before being committed. The author takes full responsibility for the correctness of the code and the accuracy of the scientific claims in the paper. AI assistance accelerated the integration and debugging phases of the project, but the underlying knowledge of the tools, the cluster, and the biology came from prior coursework and direct experience.